

**A Commitment-based Analysis of
‘Believe’ as an Implicature Trigger**

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2022

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Acknowledgements

I am most grateful to my parents who have supported me emotionally and financially so that I have had the opportunity to study in Edinburgh; and to my supervisors, Dr Wataru Uegaki and Dr Ciyang Qing, whose patient guidance and enlightenment have made this humble piece of work a possibility; and also to my friends from whom I have persistently interrogated for their intuitions about '*I believe*' utterances.

Abstract

The distribution of *believe* as a propositional attitude verb exhibits a complicated picture, which calls for a reasonable description of its strength. The observations that *believe* is typically used as a device for hedging but on other occasions does not allow unsureness motivate my proposal that the hedging sense of *believe* is realized by a generalized conversational implicature. Upon further examination, I argue that the implicature in question originates from scalar pair $\langle \textit{believe}, \textit{know} \rangle$ that is brought about by a scale of levels of commitment to the truth of the complement. Using Lauer's (2017) propositional language, I compositionally derive that *know* induces full assertoric commitment to the truth of its complement while *believe* induces a level of commitment to the truth of its complement that is lower than full assertoric commitment. This commitment-based analysis provides a plausible characterization of the strength of *believe* as it delivers desired predictions about the distribution of the predicate.

Keywords: belief, knowledge, assertion, commitment, conversational implicature.

1 Introduction

People talk about their attitudes toward the truths of propositions using different predicates with varied strengths: for example, *to know that p* is intuitively much stronger than *to suspect that p*, in that the agent is more confident that *p* is true in the case of knowledge. Some predicates of this kind, e.g., *to believe*, exhibit a quite complicated picture. In daily discourse, '*I believe*' is often used as a device for hedging, signaling the speaker's unsureness and cautiousness, as is shown in (1):

- (1) I believe I have solved the problem (but I can't be sure).

But in some contexts, the hedging sense of *believe* is blocked, as is shown in (2):

- (2) A: Tell me about your political belief.
B: I believe capitalism is better (?but I'm not sure). (Koev, 2019)

Furthermore, sometimes we use '*I believe*' to report propositions that we deem highly likely to be true, as in (3); but on other occasions we also use '*I believe*' to report something no more than an educated guess, as in (4):

- (3) I believe that the Earth spins around the Sun.
- (4) Context: In a horse race, the probabilities of winning for horse A, B and C are 45, 28, 27 % respectively.
I believe horse A will win. (Hawthorne et al., 2016)

Addressing the complicated picture of the distribution of *believe*, the goals of this dissertation are to characterize the strength of this predicate with a plausible metric, and to investigate the essence of the varied strengths of *believe* and other predicates of the kind (e.g., *know*).

The structure of the present dissertation is as follows: Section 2 is a literature review on different perspectives on the strength of *believe* and the metrics used to theorize it. Section 3 discusses the nature of the hedging sense of *believe* and proposes that it is realized by a

generalized conversational implicature with the scalar pair *<believe, know>*. Section 4 borrows Lauer's (2017) propositional language and derives the scalar pair *<believe, know>* as a scale of commitment in a compositional fashion. Section 5 discusses some further questions and the limitations of the dissertation. Section 6 concludes the positive contributions of the dissertation.

2 Literature Review

2.1 Weak belief

Hawthorne et al. (2016), among others, argue that belief is weak in contrast with assertion, which requires certainty, and in this way deny Entitlement Equality.¹ The first piece of evidence is the contrast between belief and assertion in Moore-paradoxical sentences:

- (5) a. ??It's raining, but I'm not sure it's raining.
 b. I believe it's raining, but I'm not sure it's raining.

Hawthorne et al. propose that the contrast above originates from the different strengths of belief and assertion in that in order to assert p , one needs to be sure that p ; while one can believe p even if one has not ruled out the doxastic possibility that p is false. This indicates that the standards for assertion are not the same as those for belief – the former is higher. One possible reply to this argument by Stanley (2008), as noticed by Hawthorne et al., is that the use of *believe* in sentences with the form '*I believe p but don't know p* ' is not an actual belief report, but merely cases in which one has weak reasons in support of the truth of a proposition. This view implies that there is more than one literal meaning of the lexical item *believe*, i.e., *believe* is ambiguous. It is rather implausible, as argued by Koev (2019) and Hawthorne et al. themselves, that we adopt a lexical ambiguity view of the hedging sense of *believe* which predicts the following sentence to be felicitous:

- (6) ??I believe_{weak} it's raining, but I don't believe_{strong} it's raining.

The fact that (6) is impossible to interpret suggests that *believe* is not ambiguous empirically. In addition, conceptually the two readings are logically dependent, in that the stronger sense should entail the weaker sense. Therefore, ideally we can find a more plausible theory to account for the seemingly weak strength of belief.

The second piece of evidence offered by Hawthorne et al. is that belief is as weak as a variety of states that are pre-theoretically considered weaker than belief:

¹ Entitlement Equality refers to the thesis that the evidential warrant required for asserting a proposition is just the same as the warrant for believing it. Milne (2012) and Stanley (2008) also reject Entitlement Equality.

- (7) ??Tim is of the opinion that it will rain, but he doesn't go so far as to think/believe that it will.

The fact that denying belief that p while showing some doxastic inclination toward p sounds contradictory indicates that a weak epistemic state of '*being of the opinion of p* ' is as strong as belief.² Even replacing the weak modal in (7) with *suspect* would lead to a certain extent of oddity, as observed by Hawthorne et al. However, replacing the predicate in (7) with an uncontroversially strong predicate like *know* or *be sure* completely cancels the infelicity, as observed by Mandelkern & Dorst (2021):

- (8) Tim is of the opinion that it will rain, but he doesn't go so far as to know/be sure that it will.

A further piece of evidence for weak belief discussed in Hawthorne et al. (2016) concerns with neg-raising, a linguistic phenomenon that arises with a set of predicates. For example, '*one does not believe that p* ' typically indicates that '*one believes that not- p* ', i.e., '*one disbelieves that p* '. The observation emphasized by Hawthorne et al. is that the predicates allowing neg-raising are all weak mental state verbs – for instance, neg-raising does not occur with *know*, an uncontroversially strong attitude predicate: '*John doesn't know p* ' doesn't suggest that '*John knows not- p* '. This observation further supports the thesis that belief is weak, in that it behaves more like weak predicates in terms of its compatibility with neg-raising.

The final piece of evidence proposed by Hawthorne et al. in support of a weak account for belief is that rational belief is compatible with relatively low **credence** (also known as epistemic probability). The threshold view of the relationship between belief and credence suggests that one is entitled to believe p just in case they are entitled to have a sufficiently high credence in p , and this threshold is typically considered to be 50%, i.e., one is entitled to rationally believe it if and only if they consider something to be more likely to happen than not, which is rather intuitive. However, consider for example the case of a horse race (Hawthorne et al., 2016) in which the probabilities of winning for horse A, B and C are 45, 28, 27 % respectively. In this scenario, it is perfectly natural for one to say, "*I believe horse A*

² This point is further discussed in Section 5.2.

will win”, while the agent’s credence in the proposition is well below 0.5. The relation between belief and credence will be further discussed in Section 2.3.

2.2 Strong belief

The opposite perspective is that the modal strength of *believe* is strong, and the weak reading in the hedging use is caused by other factors. For instance, Williamson (1996, 2000) endorses a knowledge norm for belief, i.e., one must only believe *p* if they know *p*. This equates *believe* with an uncontroversially strong predicate *know* in terms of their modal forces, suggesting that the standards for attributing belief are high.

An even more radical proposal explicitly equates belief with credence 1 or certainty, whose proponents include Clarke (2013) and Koev (2019).³ Koev (2019) proposes a strong, gradable semantics for *believe* and argues that the bare use of *believe* requires credence 1, and that the hedging sense of belief is not baked into the lexical entry but triggered by certain context and represented by a low level of commitment to the truth of the proposition combined with *believe* (this point will be further discussed in Section 2.4).

The first argument of Koev for the thesis that belief amounts to credence 1 is concerned with missing quantity implicatures: if the rational credence required for forming belief is lower than 1, then ‘*A believes p*’ will always implicate that ‘*A has some doubt (or not fully convinced) that p*’.⁴ However, empirically this is not the case:

- (9) a. Kamala believes that America needs universal health care.
b. Kamala is not (fully) convinced that America needs universal health care.
- (Koev, 2019)

(9a) does not implicate (9b), which, as Koev claims, is contradictory with the prediction of a weak theory of belief.

³ See also Levi (1991, 2004).

⁴ This actually does not require the strong claim that belief amounts to credence 1. An alternative explanation is discussed in Section 5.2.

Moreover, there is a contrast between how strong and weak modals interact with conjunction, in that strong modals are closed under conjunction, but weak modals are not. For example, a strong predicate like *be certain* exhibits closure under conjunction while a weak predicate *be possible* does not, as observed by Koev (2019):

- (10) I am certain that London is sunny and I am certain that Paris is raining.
 $\models$ I am certain that London is sunny and Paris is raining.

- (11) It is possible that London is sunny and it is possible that Paris is raining.
 $\not\models$ It is possible that London is sunny and Paris is raining.

Since *believe* also exhibits closure under conjunction, Koev (2019) argues that *believe* aligns with strong modals, further rebuking the idea that belief is weak:

- (12) I believe that London is sunny and I believe that Paris is raining.
 $\models$ I believe that London is sunny and Paris is raining.

It is worth noting that whether *believe* is closed under conjunction is still under debate in the philosophical literature, with detractors pointing out that the closure under conjunction of *believe* leads to the lottery paradox.⁵ Koev (2019) replies to this concern by denying the premise that one is justified to attribute belief when their credence toward a proposition is high but not maximal, further advocating the idea that belief is equivalent to credence 1.

Furthermore, Koev (2019) also disapproves Hawthorne et al.'s (2016) argument that *believe* is compatible with neg-raising so it belongs to weak predicates, claiming that the alleged link between weak modal force and neg-raising is not reliable. Under Horn's (1989: ch.5) classification of modals based on their strength, there are three types of modals: weak scalars (e.g., *be possible*), mid-scalars (e.g., *be likely*) and strong scalars (e.g., *be certain*). Horn's crosslinguistic investigation shows that compatibility with neg-raising is not a reliable indicator of modal strength; strong scalars may as well license neg-raising. Therefore, the fact that *believe* licenses neg-raising does not necessarily indicate that it is weak.

⁵ See also Logins (2020).

2.3 Against a credence-based analysis

Each of the perspectives mentioned above has its own merits, but a theory of the strength of belief that is uniformly strong or weak cannot cover the full distribution of the predicate. We need a metric for the strength of belief that makes sure that belief is not as strong as assertion, so that the hedging sense of *believe* is not excluded; and meanwhile makes sure that belief is not too weak, so that statements in the form of '*I believe p and I also believe not p*' lead to inconsistency.

Jackson (2020 & 2022) observes that the strengths of our beliefs in different propositions are not exactly the same: one's beliefs in necessary truths (Hájek, MS) like "*1+1=2*" are stronger than their beliefs in contingent truths like "*it will be sunny tomorrow*". This fact motivates philosophers to turn to credence, which is a fine-grained range of [0,1] representing the level of confidence of the agent's belief in different propositions.

It is quite intuitive to use the threshold of credence for the ascription of justified belief to theorize the modal force of *believe*: the higher the threshold, the stronger the predicate. For this very reason, the compatibility of rational belief and low credence is used by Hawthorne et al. (2016) as evidence that *believe* is weak. However, though intuitive, a credence-based analysis of the strength of *believe* may not be as plausible as it seems to be.

First, a credence-based analysis is unable to account for the full range of data related to the relative strengths of belief and assertion. On the one hand, Hawthorne et al.'s (2016) denial of Entitlement Equality suggests that the standard for assertion is something close to full confidence, and in comparison, belief only requires relatively little confidence, as is shown by the contrast in (5). Now, if this standard is represented by credence, then it would be a requirement that the credence threshold for assertion is (close to) 1. On the other hand, however, Mandelkern & Dorst's (2021) thesis that assertion is weak is supported by various scenarios and contexts where the agent can naturally assert *p* even if their credence in *p* is astonishingly low, as is shown by (13) that assertion is fully compatible with guessing contexts. Both proposals are supported by empirical data, and therefore a credence-based analysis fails to account for the full range of data.

- (13) A: What year did the Seven Years' War start?
 B: I don't know. I know it was in the 1700s, but I don't know more than that.
 A: Ok, well you're out of lifelines, so you'll just have to guess.
 B: Ok, **the war started in 1760.**

(Mandelkern & Dorst, 2021)

Furthermore, Jackson (2022) suggests that belief and credence are completely descriptively independent in that each of the doxastic attitudes of the tripartite model of belief (belief, disbelief and withholding belief) is descriptively compatible with all levels of credence in $[0,1]$. This thesis is supported by empirical data including contexts where belief is compatible with very low and even close to 0 credence (e.g., the infinite preface scenarios (Smith 2016), and the dart and bullseye case discussed in Moss (2018)). A credence-based analysis would now predict that *believe* is as weak as *be possible*, since any level of doxastic possibility can lead to belief.⁶ However, empirical data goes against this prediction, as we can see the contrast below:

- (14) a. ??I believe it is raining, and I also believe it is not raining.
 b. It is possible that it is raining, and it is also possible that it is not raining.

The contradictions induced by the predictions of a credence-based analysis and the corresponding empirical data suggest that credence fails to characterize the strength of *believe* and thus we need an alternative metric to fully account for the distribution of the predicate.

2.4 Commitment: an alternative metric

Although Koev's (2019) claim that belief amounts to credence 1 is rendered implausible by the empirical data supporting the complete descriptive independence of belief and credence, his idea of turning to the notion of commitment to explain the hedging use of *believe* provides an alternative metric for theorizing the strength of *believe*. Koev invokes the standard anti-presupposition theory of Percus (2006), arguing that *believe* and *know* constitutes a pair of p(presuppositional)-alternatives $\langle \textit{believe}, \textit{know} \rangle$ because *know* is a factive counterpart of

⁶ A commitment-based analysis predicts the contrary. See Section 4.1.

believe and adds a presupposition to the effect that its complement is true, facilitating a scalar inference in terms of presuppositional force. In certain contexts, the presuppositionally stronger alternative is excluded, yielding the enriched meaning that ‘*believe p and not know p*’, indicating that the agent is not publicly committed to the truth of *p*.⁷

While Koev’s theory gives us a clue to the nature of the hedging sense of *believe* as a scalar implicature, it characterizes the strength of *believe* based on credence and holds on to the strong claim that belief amounts to credence 1, which is implausible in light of the empirical data showing that belief and credence are descriptively independent.

Lauer (2017) as well proposes a commitment-based account of the declarative force of *believe* with particular attention to the connection and distinction between the assertion of ‘*believe that p*’ and the assertion of *p*. Adopting Spohn’s (1988, 1990, 2012) ranking theory, Lauer develops a theory of the strength of *believe* represented by graded commitments in a compositional fashion that accounts for the four desiderata that are in the center of debate on the topic: 1) assertion of ‘*believe that p*’ induces weaker commitment than direct assertion of *p*; 2) Moore-paradoxical sentences like ‘*p, but I don’t believe p*’ are inconsistent in terms of commitments; 3) assertions of ‘*I believe p*’ and ‘*I believe ¬p*’ are jointly inconsistent; 4) assertions of *believe* are closed under conjunction. Lauer’s theory to a certain extent reconciles the weak theory and the strong theory of belief by accounting for a wide range of data including Moore-paradoxical sentences and hedging use of *believe* with a unified analysis of *believe*. In this regard, commitment is arguably a suitable metric to measure the strength of *believe*.

Note, however, Lauer’s theory does not directly provide an account for the nature of the hedging use of *believe*. It is not yet clear how expressing a certain level of commitment to the truth of a proposition typically implies the denial of a higher level of commitment. But if we integrate this theory with the scalar implicature account inspired by Koev (2019), we can try to represent the scalar pair <*believe, know*> as a scale of commitment and thus account for the hedging use of *believe* while theorizing the relative strengths of *believe* and *know*.

⁷ There are different interpretations for the concept of commitment. Tebben’s (2018) interpretation is the closest one for the purpose of the present dissertation: *if one is committed to the truth of p, taking not-p as a premise reveals an objectionable kind of incoherence, one that indicates irrationality in either one’s commitments or one’s reasoning, and so is impermissible*. As we will see in the following sections, ‘*believe that p*’ does induce a certain level of commitment to the truth of *p*.

In summary, the divided perspectives along with the related empirical data indicate a complicated picture regarding the issue of characterizing the strength of *believe*. A credence-based analysis is proved unable to solve the problem as it cannot account for the full range of data that is in the center of debate. A structure of different levels of commitment, on the other hand, appears to be a more suitable representation of belief. Based on this, in the following sections I will discuss the nature of the hedging sense of *believe* by integrating Lauer's (2017) commitment-based analysis of *believe* and the scalar implicature account inspired by Koev (2019), as it sheds light on the relative strength of *believe* compared to other predicates of the kind (e.g., *know*) and thus the essence of the varied strengths and distributions of these predicates.

3 ‘I believe’ as a hedging device

3.1 A generalized conversational implicature analysis

Believe is often used in utterances as a device for hedging: when a speaker utters “*I believe that p*”, they convey that they are not entirely sure that *p* is true (Lauer, 2017). In other words, assertion of ‘*believe that p*’ typically indicates a lower commitment to the truth of *p* compared to direct assertion of *p* or, say, assertion of ‘*know that p*’, as is well demonstrated by Hawthorne et al.’s (2016) denial of Entitlement Equality. As a matter of fact, we normally and automatically infer the agent of ‘*believe that p*’ is not entirely sure that *p* – this is in line with Grice’s (1975) and Levinson’s (2000) characterization of generalized conversational implicature.

One property of the hedging use of *believe* that serves as evidence in support of a conversational implicature theory is that the hedging sense is cancellable. One could naturally cancel the hedging sense either explicitly, as shown in (15), or implicitly by context, as shown in (16):

(15) I believe it is raining. In fact, I know it is raining.

(16) A: Tell us about your political beliefs.

B: I believe capitalism is better than socialism (?but I’m not sure).

(Koev, 2019)

In (15), the implicature is explicitly cancelled by the speaker’s following up with a stronger predicate *know*. This is reminiscent of a typical scalar implicature and sheds light on the source of the unsureness conveyed by the speaker of “*I believe that p*”: the speaker implicates the denial of a proposition stronger than the one said. Just as saying “*some athletes smoke*” implicates that “*not all athletes smoke*”, by saying “*I believe that p*” the speaker also implicates that “*it is not the case that I know that p*”, yielding the hedging sense.

In (16), the utterance of A presupposes that the addressee has and knows their political beliefs. By participating in the conversation, B accommodates the presuppositions; and subsequently it becomes common ground that B knows their political beliefs, and the

implicature is therefore cancelled, which explains why the hedging sense is inaccessible in such a context.⁸

In this sub-section, I proposed that the hedging sense of *believe* is a generalized conversational implicature based on the data showing its cancellability, and that the unsureness conveyed by the speaker of “*I believe that p*” is caused by the implicated denial of “*I know that p*”, a stronger alternative to the proposition uttered. The next section will speculate the nature of this implicature.

3.2 The nature of the implicature

It is uncontroversial that as a factive verb, *know* is stronger than *believe* as the agent of the former considers the proposition combined with the predicate to be true. However, how this factive part of *know* is realized, and thus in what way *know* is stronger than *believe*, is an open issue still debated in the literature.

On the one hand, some advocate the idea that since factive verbs (e.g., *know*, *regret*) are widely agreed to be presupposition triggers, *to know that p* **presupposes** *p is true* (Kiparsky & Kiparsky, 1970 via Beaver et al., 2021). This theory is supported by the fact that the truth of the complement of *know* still holds in embeddings, which is the hallmark of presuppositions:

- (17) a. Jack knows that 51 is prime.
- b. 51 is prime.
- c. Jack doesn’t know that 51 is prime. (negation)
- d. If Jack knows that 51 is prime, he would be very surprised. (antecedent of conditional)
- e. Does Jack know that 51 is prime? (question)

For (17c-e) which embed (17a) under various operators, (17b) always holds, indicating that *to know that p* presupposes that *p is true*. In this way, as the factive counterpart of *believe*, *know* is presuppositionally stronger (Koev 2019), which is analogous to Heim’s (1991) proposal

⁸ Technically, in this case the implicature never arises in the first place. Rather, the possibility of the implicature in question is eliminated by the context implicitly.

that the definite article *the* is presuppositionally stronger than the indefinite article *a(n)*. Pairs like $\langle a(n), the \rangle$ and $\langle believe, know \rangle$ are thus called presupposition-based scales (p-scales) in Percus (2006), among others.⁹ Crucially, Percus proposes the notion of antipresupposition: the utterance of a weaker lexical item in a p-scale implies that the presupposition of its presuppositionally stronger counterpart is not certain to hold. Based on this, Koev (2019) interprets the hedging sense of *believe* as a realization of antipresupposition:

- (18) a. Actual utterance: John believes it's raining.
 b. Alternative utterance: John knows it's raining.
 c. Antipresupposition: It's not certain that it's raining.

On the other hand, however, some also support the idea that *to know that p* **entails** that *p* is true. The proponents suggest that *know* is of the class of words that trigger conventional implicatures by means of entailments but not presuppositions, just like *start* and *stop* applied to gerunds like *cheating* (Davis, 2019; Simons, 2001: 432-3). In fact, while Heim (1991) argues that the indefinite and definite articles form a p-scale pair, Hawkins (1991) following Russell (1905) argues that $\langle a(n), the \rangle$ is an entailment-based scale pair (e-scale), in which the non-uniqueness of the indefinite article is derived as the result of a scalar implicature as *the* entails the unique existence of an object. Analogously, if we subscribe to the entailment theory of the factive part of *know*, we can as well derive $\langle believe, know \rangle$ as an e-scale pair.

Crucially, both theories lead to the inference that $\langle believe, know \rangle$ is a scalar pair. It then seems plausible to establish a link between the notions of p-scale and e-scale and the proposal of the present dissertation: the previous sub-section argues that the hedging sense of *believe* is delivered by a generalized conversational implicature that denies “*I know that p*”, a stronger proposition than “*I believe that p*”; this can exactly be derived if we treat $\langle believe, know \rangle$ as a scalar pair. In this way, we yield a scalar pair $\langle believe, know \rangle$ that is analogous to a typical scalar pair of quantifiers like $\langle some, all \rangle$, either as a p-scale pair or an e-scale pair. For the purpose of convenience, I will treat $\langle believe, know \rangle$ as an e-scale in the following sections, as entailments fit directly in with Lauer's (2017) formal system which is crucial for representing the scalar implicature in question as a scale of commitment.

⁹ See also Schlenker (2012) and Singh (2011).

3.3 Towards a scale of commitment

It is obvious that quantifier scalar pairs are realized by varied quantities denoted by individual quantifiers. However, it is not as clear by which metric of strength the pair of *<believe, know>* is realized. In other words, it is a matter of how to interpret the effect of the factive entailment of *know*: when someone asserts ‘*I know that p*’, does the entailment ‘*p is true*’ induce a higher credence in *p* or a higher commitment to *p* than asserting the weaker alternative ‘*I believe that p*’?

Empirical data suggests a picture that goes against a credence-based analysis of *believe*. Recall the discussion in Section 2.3 that credence is unlikely to be the defining metric of the strength of *believe* in that 1) Jackson (2022) proposes that belief and credence are completely descriptively independent based on empirical data showing the compatibility of the ascription of belief with full range of credence [0,1]; 2) if the credence threshold defines the modal strength of *believe*, Hawthorne et al.’s (2016) denial of Entitlement Equality would imply that one is entitled to assert *p* only if their credence in *p* is something close to 1; this is in conflict with Mandelkern & Dorst’s (2021) observation that assertion is compatible with credence levels that are quite low – both proposals are backed by empirical data, and therefore a credence-based analysis fails to account for the full range of data.

Moreover, a credence-based account for the strength of *know* can also be implausible. Holguín (2019) observes that in daily discourse, one is justified to say ‘*I know that p*’ even if their credence is lower than 0.5 depending on the specific contexts. This implies that the quantitative relation of the credence thresholds for *know* and *believe* cannot be fixed – either one could be stronger depending on the contexts. Consequently, it leads to conflict with some widely accepted principles about knowledge: if *S* knows that *p*, then *S* believes that *p* and is justified to believe that *p*; that is, *to know that p* always appears to be stronger than *to believe that p*. Therefore, a credence-based analysis could fail to predict the relation of the strengths of *know* and *believe* and therefore is unable to explain why the scalar pair *<believe, know>* exists.

On the other hand, a commitment-based analysis proposed by Lauer (2017) suggests that the assertion of ‘*believe that p*’ appears to be weaker than the assertion of *p* because the assertion of ‘*believe that p*’ does not induce full assertoric commitment to *p*. That is to say, the

standard for asserting '*believe that p*' is lower than that for directly asserting *p* in terms of the level of commitment to the truth of *p* required from the speaker, which is in line with Hawthorne et al.'s (2016) denial of Entitlement Equality. Crucially, this analysis does not conflict with Mandelkern & Dorst's (2021) thesis that assertion is weak – assertion can be weak in terms of the requirement for credence level, but it is stronger than belief in terms of commitment. In this way, a commitment-based analysis is able to cover the full range of data in both Hawthorne et al. and Mandelkern and Dorst that a credence-based analysis fails to account for.

Now that we have a plausible metric, commitment, to characterize the strength of *believe* in utterances, the next step is to use this very metric to speculate the strength of *know*. In this way, we are able to interpret the factive entailment of the assertion of '*know that p*' as inducing a higher commitment than the assertion of '*believe that p*', and thus the entailment-based scalar pair $\langle \textit{believe}, \textit{know} \rangle$ is represented by a scale in commitment. But how can we measure the level of commitment to the truth of *p* that is induced by the assertion of '*know that p*'?

Intuitively, the relative levels of commitment induced by the assertions of '*believe that p*' and '*know that p*' can be demonstrated by the fact that the latter can be used to strengthen the former, but not vice versa:

- (19) a. I believe that it is raining. In fact, I know it is raining.
- b. # I know that it is raining. In fact, I believe it is raining.

This indicates the commitment induced by the assertion of '*know that p*' is higher than that induced by the assertion of '*believe that p*'. Analogously, we may postulate the relative levels of commitment induced by the assertion of '*know that p*' and the assertion of *p*:

- (20) a. # It is raining. In fact, I know it is raining.
- b. # I know it is raining. In fact, it is raining.

The sentences in (20) are rather uninformative, which indicates that the assertion of '*know that p*' and the assertion of *p* cannot strengthen each other. Furthermore, it is natural to ask

pushback questions with ‘*know*’ after assertions, also indicating the equal strength of the assertion of ‘*know that p*’ and the assertion of *p*:

- (21) A: It is raining.
B: How do you know that?

Based on these data, we are in a position to hypothesize that the assertion of ‘*know that p*’ induces full assertoric commitment to the truth of *p*, forming a scale of commitment with the assertion of ‘*believe that p*’ and thus the e-scalar pair <*believe, know*>. The next section borrows Lauer’s (2017) propositional language and formal mechanisms and aims to derive this hypothesis in a compositional fashion.

4 Deriving the scale

4.1 Lauer's (2017) propositional language

Lauer's (2017) system is based on the ranking theory of Spohn (2012), representing graded beliefs with the help of two functions:

- a) **Ranking mass function** (Spohn 2012: 70): Given a set of worlds W , a ranking mass function is any function $k : W \rightarrow (\mathbb{N} \cup \{\infty\})$ such that $k^{-1}(0) \neq \emptyset$.
- b) **Ranked belief function** (Spohn 2012: 75): Given a ranking mass function k , the corresponding ranked belief function β_k is that function $\wp(W) \rightarrow (\mathbb{N} \cup \{\infty\})$ such that i) $\beta_k(W) = \infty$; ii) $\beta_k(\emptyset) = 0$; iii) for all non-empty $A \subseteq W$: $\beta_k(A) = \min\{k(w) \mid w \in A\}$. A ranked belief function maps each proposition to a measure of its belief, so the higher the rank of a proposition is, the more it is believed.

The object language proposed by Lauer is a simple propositional language enriched by some modal operators for belief:

- a) **Syntax**¹⁰:

Apart from the clauses of the universal definition of all propositional languages (propositional letters, negation of formulas, etc.), the language $\mathcal{L}_{P,I}$ (let P, I be disjoint sets of proposition letters and individuals respectively) is enriched by belief formulas: If $\phi \in \mathcal{L}_{P,I}$ and $i \in I$, then $(Bel_i(\phi)) \in \mathcal{L}_{P,I}$, where $Bel_i(\cdot)$ is an operator representing the content of *I believe* statements.

- b) **Semantics**:

Models: models are possible-worlds ones with two extra elements to interpret the belief operator: a function K specifying the belief state for each agent at each world, and a threshold for belief ascription b . Thus, a model for $\mathcal{L}_{P,I}$ is $M = \langle W, V, K, b \rangle$ such that i) W is a set of possible worlds; ii) $V: P \rightarrow \wp(W)$ is a valuation for the

¹⁰ Entailments can directly fit in with the system, but additional rules can be added to the syntax of the propositional language to make it compatible with presuppositions. In this way, *<believe, know>* as a p-scale can also be represented as a commitment scale.

propositional letters; iii) $K: I \times W \rightarrow \mathbb{K}_W$ is a function that maps each pair of individual-world to a ranking mass function; iv) $0 < b \in \mathbb{N}$.

Denotation function: given a model $M = \langle W, V, K, b \rangle$, the denotation function $\llbracket \cdot \rrbracket^M: \mathcal{L}_{P,I} \rightarrow \wp(W)$ satisfies that (omitting the model parameter): i) $\llbracket p \rrbracket = V(p)$ for all $p \in P$; ii) $\llbracket \neg \phi \rrbracket = W - \llbracket \phi \rrbracket$; iii) $\llbracket \phi \wedge \psi \rrbracket = \llbracket \phi \rrbracket \cap \llbracket \psi \rrbracket$; iv) $\llbracket Bel_i(\phi) \rrbracket = \{w \in W \mid \beta_{K(i,w)}(\phi) \geq b\}$.

c) **Pragmatics:**

The commitments of an agent are represented as constraints on ranking functions.

Commitment frames: A commitment frame for $\mathcal{L}_{P,I}$ is any $\mathcal{C}_i = \langle M, i, a \rangle$, where:

i) $M = \langle W, V, K, b \rangle$ is a model for $\mathcal{L}_{P,I}$; ii) $i \in I$ is an individual; iii) $b < a \in \mathbb{N}$ is the threshold for assertion.

Commitment states: Given a commitment frame $\mathcal{C}_i = \langle M, i, a \rangle$, a commitment state C_i is partial truth function of ranking mass functions such that $C_i(k)$ is defined iff k is admissible for i in M .¹¹

Declarative update: Given a commitment frame $\mathcal{C}_i = \langle M, i, a \rangle$, the update function $+$ is the following function from commitment states and formulas to commitment states:

$$C + \phi = \lambda k. C(k) \ \& \ \beta_k(\llbracket \phi \rrbracket) \geq a$$

Belief update: Given a commitment frame $\mathcal{C}_i = \langle M, i, a \rangle$ with a model $M = \langle W, V, K, b \rangle$, for any commitment state C_i and formula ϕ :

$$\text{For any } k: \text{ if } [C_i + Bel_i(\phi)](k) = 1, \text{ then } k(\llbracket \phi \rrbracket) \geq b$$

Conjunction: for any commitment state C_i and formula ϕ, ψ : $C_i + (\phi \wedge \psi) = C_i + \phi + \psi$.

The crucial result derived from Lauer's system is the commitment level of the assertion of '*believe that p*' compared to that of the assertion of p : declarative update with '*believe that p*' (i.e., belief update) does not induce full assertoric commitment (which is represented by threshold a) to the truth of p , but does induce a commitment to the truth of p to the level of b , which is lower than a . The system also proves that the assertions of '*believe that p*' and '*believe that not p*' jointly lead to inconsistency in commitment.

¹¹ Given a language $\mathcal{L}_{P,I}$, model $M = \langle W, V, K, b \rangle$, a pointwise ranking mass function k is admissible for $i \in I$ in M iff $\forall v \in k^{-1}(0) : K(i, v) = k$.

With the propositional language above, we are now able to derive the commitment level induced by the assertion of ‘*know that p*’.

4.2 Scalar pair <believe, know> as commitment scale <b, a>

Consider <*a(n)*, *the*> as an entailment-based scalar pair. Aside from the additional entailment of uniqueness of the stronger alternative *the*, the two lexical items in the pair share the same entailment of existence (Hawkins, 1991). Analogously, for entailment-based scale <*believe*, *know*>, aside from the additional entailment of *know* that its complement is true, the two lexical items in the pair can be assumed to be truth-conditionally equivalent. In other words, *to know that p* is composed of two parts: the speaker’s belief that *p* and the entailment that *p* is true. Consequently, the assertion of ‘*know that p*’ is equivalent to the assertion of ‘(*believe that p*) and (*p*)’, stating the entailment explicitly.¹²

Our hypothesis is that the assertion of ‘*know that p*’ induces full assertoric commitment to the truth of *p*. That is to say, the threshold for the assertion of ‘*know that p*’ is the same as the threshold for the assertion of *p* which is *a*. To prove this within Lauer’s (2017) system is to prove that given a commitment frame $\mathcal{C}_i = \langle M, i, a \rangle$ with a model $M = \langle W, V, K, b \rangle$:

For any k , if $[C_i + \text{Know}_i(\phi)](k) = 1$, then $k(\llbracket \phi \rrbracket) \geq a$

Where $\text{Know}_i(\cdot)$ is an operator representing the content of ‘*I know*’ statements

Proof. Following our assumption regarding entailment-based scales, $\text{Know}_i(\phi) = (\text{Bel}_i(\phi) \wedge \phi)$. Then $C_i + \text{Know}_i(\phi) = C_i + (\text{Bel}_i(\phi) \wedge \phi)$. By **Conjunction**, $C_i + (\text{Bel}_i(\phi) \wedge \phi) = C_i + \text{Bel}_i(\phi) + \phi$, which indicates that two updates are involved: one belief update and one declarative update. Now, for any k , if $[C_i + \text{Bel}_i(\phi) + \phi](k) = 1$, both updates must hold to be true. In derivational terms, the belief update requires that for any k , if $[C_i + \text{Bel}_i(\phi)](k) = 1$, then we have $k(\llbracket \phi \rrbracket) \geq b$ (by **Belief update**) and an updated commitment state¹³; the declarative update then requires that for any k , $[C_i + \phi](k) = 1$, then we have $k(\llbracket \phi \rrbracket) \geq a$ (by **Declarative update**). Thus, for any k , if $[C_i + \text{Know}_i(\phi)](k) = 1$, then $k(\llbracket \phi \rrbracket) \geq \max \{a, b\}$. By definition of commitment frames, $b < a \in \mathbb{N}$. Therefore, for any k , if $[C_i + \text{Know}_i(\phi)](k) = 1$, $k(\llbracket \phi \rrbracket) \geq a$. $\square$

¹² Note that this is just a simplifying assumption for the purposes of this dissertation.

¹³ The order of the two updates makes no difference, as C_i stands for any commitment state.

Now, we have that the assertion of '*know that p*' has the same commitment threshold *a* as the assertion of *p* which is higher than the threshold for the assertion of '*believe that p*'. Thus, the entailment-based scale pair *<believe, know>* is represented by a commitment scale *<b, a>*: compared to its stronger alternative '*I know that p*', the assertion of '*I believe that p*' induces lower commitment to the truth of *p*.

Subsequently, the implicature can be derived with a paradigm application of Gricean theory: since the speaker has used a weaker expression (i.e., "*I believe that p*") that induces a lower commitment (i.e., *b*) to *p* where a stronger one (i.e., "*I know that p*") that induces a higher commitment (i.e., *a*) to *p* is available, they would contradict Grice's Quantity Maxim if the stronger statement that induces a higher commitment held. Consequently, they consider the stronger statement not to hold. Furthermore, they have done nothing to stop the addressee thinking that they consider so, therefore they conversationally implicate that '*It is not the case that I know that p*'.

5 Discussion

5.1 Stipulating the commitment thresholds of other predicates

In the literature discussed in Section 2, several predicates that are closely related to *believe* are used to speculate the strength of *believe*, including *be certain/sure*, *be convinced*, *be of the opinion*, etc.. One of the conflicts between the weak theory and the strong theory is that on the one hand, as Hawthorne et al. (2016) claim: the assertion of ‘*be of the opinion that p*’ and the assertion of ‘*not believe that p*’ jointly lead to inconsistency as in (7), indicating that *believe* is no stronger than *be of the opinion*; on the other hand, as Koev (2019) claims: the assertion of ‘*believe that p*’ does not induce the implicature ‘*not (fully) convinced that p*’, indicating that belief amounts to full credence and thus *believe* is strong.

A commitment-based framework helps reconcile this conflict and at the same time helps stipulate the commitment levels of the predicates in question. On the one hand, there is evidence showing that *believe* and *be of the opinion* may induce the same level of commitment, as we can see that the sentences in (22) are rather uninformative:

- (22) a. #I am of the opinion that it is raining. In fact, I believe it is raining.
b. #I believe it is raining. In fact, I am of the opinion that it is raining.

This indicates that *be of the opinion* has the same commitment threshold *b* as *believe*, which explains why utterances like (7) sound contradictory.

On the other hand, we can use the same way to speculate the commitment level induced by ‘*be fully convinced*’, as it can strengthen *believe* and be strengthened by *know*:

- (23) a. I believe it is raining. In fact, I am fully convinced that it is raining.
b. I am fully convinced that it is raining. In fact, I know it is raining.

This indicates that coarsely, *be fully convinced* induces a level of commitment that is higher than *believe* but lower than *know*. Let’s stipulate this threshold as *c*, then we have $0 < b < c < a \in \mathbb{N}$. We now have an explanation alternative to Koev’s claim regarding why ‘*believe*

that p’ does not lead to the implicature ‘*not fully convinced that p*’: the prediction of the existence of this implicature is an example of overgeneration of the Gricean schema.

Taking scalar pair <*some, all*> as an instance, there are countless statements between “*Some athletes smoke*” and “*All athletes smoke*”, but the former only implicates the negation of the latter, not the negation of any statements in between like “*13% of all athletes smoke*”, although the Gricean reasoning can be performed with any statement stronger than “*Some athletes smoke*”. Analogously, “*I believe that p*” (commitment threshold b) implicates the negation of “*I know that p*” (commitment threshold a), but not the negation of any statements in between including “*I am fully convinced that p*” (commitment threshold c), as $b < c < a$.¹⁴ In other words, *believe* and *be fully convinced* are not scalemates like <*believe, know*>. In this way, the non-existence of the implicature in question is explained without resorting to the strong claim that belief amounts to full credence.

Furthermore, the analysis that *believe* and *be fully convinced* are not scalemates also fits in well with Jackson’s (2022) thesis that belief and credence are descriptively independent: the ascription of belief is not governed by a threshold of credence, *believe* and *be fully convinced* would not form a scale in credence, even if *be fully convinced* encodes full subjective probability, i.e., credence, as Koev (2019) claims.

5.2 Epistemic Posturing and the commitment threshold of assertion

Epistemic Posturing refers to the norm of assertion proposed by Mandelkern & Dorst (2021) that “*in performing the speech act of asserting p, act as if you are absolutely certain of p*”. The empirical data that motivate them to propose such a strong norm for assertion include Moore-paradoxical sentences with attitude predicates that are pre-theoretically stronger than *know*. It appears that the assertion of *p* and the denial of absolute certainty of *p* jointly lead to inconsistency:

¹⁴ A potential question is why the assertion of ‘*be convinced that p*’ does not implicate the denial of ‘*know that p*’. In other words, why *be convinced* and *know* do not form a scalar pair like <*believe, know*>, as there is a scale in commitment <c, a>. A possible answer is that the lexical semantics of *be convinced* involves a change in doxastic state which *know* simply lacks, so even apart from the factive entailment/presupposition, *know* and *be convinced* cannot be truth-conditionally equivalent.

- (24)
- a. #John will bring Indian, but I'm not completely confident that he will.
 - b. #John will bring Indian, but there's the tiniest chance that he won't.
 - c. #Miriam lost, but I wouldn't bet my life that she lost.
 - d. #Slippery Pete's going to win this, but I'm not absolutely certain he will.
 - e. #The butternut squash are in aisle 4, but I can't absolutely, infallibly rule out every possibility in which they aren't.

(Mandelkern & Dorst, 2021)

Recall the proposal of the current dissertation that the assertion of '*know that p*' induces the same level of commitment as the direct assertion of *p* which is represented by threshold a. Consequently, predicates that are stronger than *know* are supposed to induce a higher level of commitment than direct assertion. Now, this becomes problematic because our analysis predicts that expressing a certain level of commitment to the truth of a proposition and subsequently denying a higher level of commitment should not lead to inconsistency. Therefore, assuming that the attitude predicates in (24) like *be absolutely certain* and *bet one's life* are indeed stronger than *know*, the sentences in (24) should be felicitous. However, empirical data suggests the contrary.

A possible solution is to examine these intuitively stronger-than-*know* predicates more closely. If they are proved not to be stronger than *know*, then the infelicity of the sentences in (24) is explained. However, speaker's judgements vary in terms of the informativeness of the sentences involved in strengthening diagnostics as shown in (25), possibly because epistemic states like '*bet my life that p*' seldom appear in our daily life:

- (25)
- a. I know John will bring Indian. In fact, I'm absolutely certain that he will.
 - b. I know Miriam lost. In fact, I bet my life that she lost.
 - c. I know Kamran's going to win this. In fact, I'm completely confident that he will.

Therefore, the relation of the strengths of *know* and predicates like '*be absolutely certain*' and '*bet one's life*' in a structure of commitment remains an open issue for further research. If these predicates and *know* were proved equally strong, the analysis of the current dissertation could account for the infelicity of the sentences in (24). If these predicates were indeed

proved to be stronger than *know*, it would result in a potential limitation of the analysis of the current dissertation.

In summary, for future research it is worth investigating how various attitude predicates fit in with the commitment-based analysis adopted in this dissertation. In Section 5.1, the threshold of commitment of *be convinced* is only coarsely stipulated to be in between *believe* and *know* but not compositionally derived using the formal mechanisms as in Section 4; in Section 5.2, we saw that the levels of commitment induced by some predicates cannot even be coarsely stipulated. The empirical data associated with these predicates could in turn be used for an evaluation of the analysis in this dissertation.

6 Conclusion

In this dissertation, I have examined two relevant metrics discussed in the literature: credence and commitment, aiming to characterize the strength of *believe*. I have argued that a credence-based analysis fails to account for the full distribution of *believe*, as empirical data suggests that belief and credence are descriptively independent, i.e., speakers' use of *believe* is not governed by a threshold of credence. Moving towards a commitment-based analysis, I first proved that the hedging sense of *believe* is realized by a generalized conversational implicature, and then showed that this implicature originates from scalar pair $\langle \textit{believe}, \textit{know} \rangle$ which is represented by a scale in commitment. Subsequently, I derived the scale of commitment in a compositional fashion with Lauer's (2017) propositional language.

The commitment-based analysis in this dissertation shows that on the one hand, compared to its factive counterpart *know*, *believe* is relatively weak in terms of the induced commitment to the truth of its complement – this makes *believe* a conversational implicature trigger, accounting for the fact that *believe* is typically used as a device for hedging and can also appear to be strong when the implicature is explicitly or implicitly cancelled. This point amounts to the denial of a knowledge norm for belief proposed by Williamson (1996, 2000), as the standard for belief ascription is lower than that for knowledge; and the denial of Entitlement Equality, as the commitment threshold for the assertion of '*believe that p*' is lower than the full assertoric commitment to *p*. On the other hand, the assertion of '*believe that p*' does induce a certain level of commitment to the truth of *p*, so that for any proposition *p*, the two statements '*believe that p*' and '*believe that not p*' jointly lead to inconsistency, ensuring that *believe* is not too weak. In this way, the current analysis captures the full range of distribution of *believe*, providing a plausible characterization of the strength of the predicate.

think believe know are all defined on the same dimension?
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